

AIGR Deep Research Report: Rat HMGCS2 (P22791) — Mevalonate/Isoprenoid Function Assignment

Gene: Hmgcs2 (rat, *Rattus norvegicus*, NCBITaxon:10116) **UniProt:** P22791 **Focus type:** function_assignment **Hypothesis slug:** mevalonate-and-isoprenoid-synthesis **GO claims under evaluation:** GO:0010142 (farnesyl diphosphate biosynthetic process, mevalonate pathway) and GO:0008299 (isoprenoid biosynthetic process)

Summary

The seed hypothesis asks whether rat HMGCS2 participates in **farnesyl diphosphate biosynthesis via mevalonate (GO:0010142)** and **isoprenoid biosynthesis (GO:0008299)** as an endogenous function, or whether any such capacity is merely ectopic (observed only in complementation or cancer-overexpression contexts). After three iterations of investigation against primary literature and curated database records, the answer is clear: **the mevalonate/isoprenoid biological-process assignment is refuted / over-annotated for HMGCS2**. Rat HMGCS2 (P22791) is the **mitochondrial-matrix, ketogenic** isoform of HMG-CoA synthase (EC 2.3.3.10). Its endogenous role is to catalyze the first committed, rate-limiting step of **ketone-body synthesis** — condensing acetyl-CoA and acetoacetyl-CoA into HMG-CoA — after which mitochondrial HMG-CoA lyase cleaves that HMG-CoA to acetoacetate and β -hydroxybutyrate. It does not feed the mevalonate/isoprenoid pathway.

The two GO claims must be assessed separately, as the hypothesis instructs. **GO:0010142** is present on HMGCS2 only as an **IBA (Inferred from Biological Ancestor) annotation from GO_Central** — a phylogenetic carry-over from the shared HMG-CoA synthase ancestral node — and is not supported by any HMGCS2-specific experimental evidence. **GO:0008299 is not currently annotated on HMGCS2 at all and should not be added**. The mevalonate/isoprenoid program legitimately belongs to the cytosolic paralog **HMGCS1**, whose downstream partner enzyme HMGCR (HMG-CoA reductase) is an integral endoplasmic-reticulum (ER) membrane protein physically inaccessible to the mitochondrial matrix.

The one genuine, retained function shared by both paralogs is the **molecular function** hydroxymethylglutaryl-CoA synthase activity (GO:0004421, IDA). The biological process that reflects HMGCS2's true endogenous role is **ketone body biosynthetic process (GO:0046951)**, and its cellular component is the **mitochondrial matrix (GO:0005759, IDA:RGD)**. The critical conceptual caveat is that HMGCS2 genuinely produces HMG-CoA — the same metabolite that seeds the mevalonate pathway — which is precisely why automated phylogenetic pipelines mis-assign the mevalonate BP term. But producing the shared metabolite in the wrong subcellular compartment is not the same as participating in the pathway. Mitochondrial localization does not settle the pathway in favor of mevalonate; it settles it against.

Key Findings

Finding 1 — HMGCS2 is the mitochondrial, ketogenic isoform; the mevalonate/isoprenoid role belongs to cytosolic HMGCS1

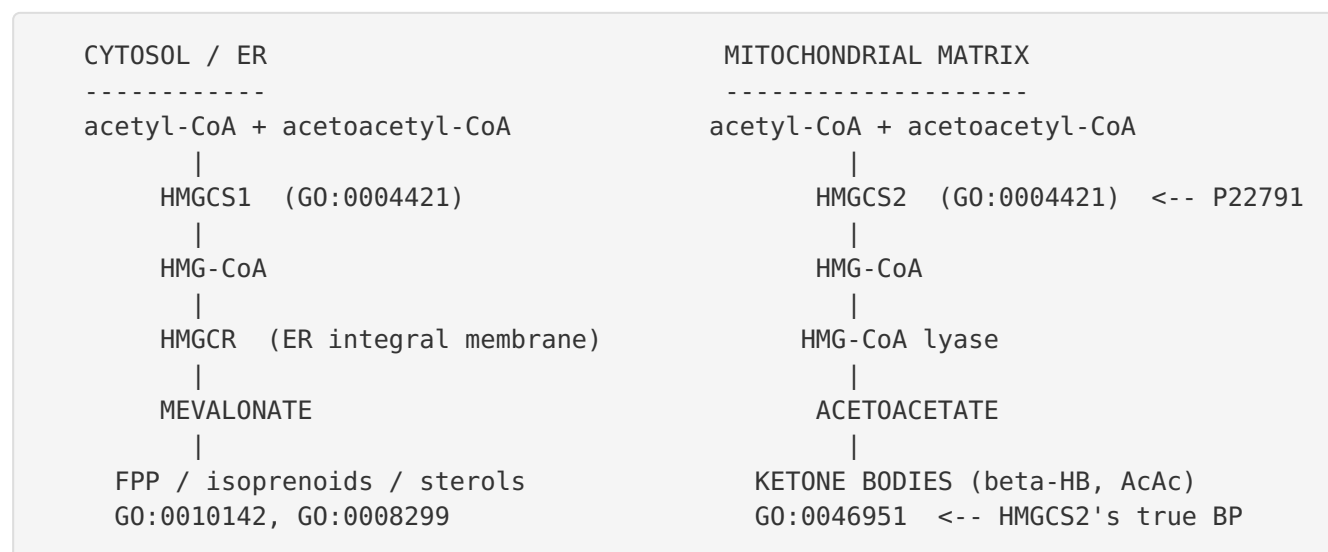
Rat HMGCS2 (UniProt P22791) is annotated with EC 2.3.3.10, a subcellular location of mitochondrion / mitochondrial matrix (GO:0005759, IDA:RGD), and a UniProt function statement describing it as the enzyme that catalyzes "the first irreversible step in ketogenesis." Across the recent primary literature — spanning kidney, liver, and placenta — HMGCS2 is uniformly and explicitly described as **the rate-limiting enzyme of ketogenesis**, and never as a mevalonate or isoprenoid enzyme. A representative renal study states that "the rate-limiting enzyme for ketogenesis, mitochondrial hydroxymethylglutaryl-CoA synthase 2 (HMGCS2), is induced in the proximal tubule of the kidney during fasting" (PMID: 41568909). The same characterization recurs independently in hepatic MASLD-MASH flux studies (PMID: 40272888, PMID: 39464122), in diabetic kidney disease (PMID: 41580118), in exercise/cognition work (PMID: 39808588), and in placental ketone delivery (PMID: 40288162).

In direct contrast, the **mevalonate committed step is assigned to the paralog HMGCS1**. A *C. elegans* aging-metabolism study identifies the worm HMG-CoA synthase HMGS-1 as "the ortholog of human HMGCS1 enzyme, which mediates the first committed step of the mevalonate pathway" (PMID: 25187565). This establishes the functional division of labor between the two paralogs: **cytosolic HMGCS1 for mevalonate/sterol/isoprenoid synthesis, mitochondrial HMGCS2 for ketogenesis**. This paralog split is the single most decisive discriminating fact for this review, because it explains both why the two enzymes share a molecular function and why only one of them carries the mevalonate biological process.

Finding 2 — Compartmentalization blocks mitochondrial HMG-CoA from feeding the cytosolic/ER mevalonate-isoprenoid pathway

Even though HMGCS2 chemically produces HMG-CoA, it cannot physiologically participate in mevalonate/isoprenoid synthesis, because **the obligatory next enzyme in that pathway, HMG-CoA reductase (HMGCR), is an integral ER membrane protein**, physically separated from the mitochondrial matrix where HMGCS2 resides. A review of sterol and nonsterol isoprenoid synthesis describes "Endoplasmic reticulum (ER)-localized 3-hydroxy-3-methylglutaryl coenzyme A (HMG CoA) reductase (HMGCR), the rate limiting enzyme in the mevalonate pathway" (PMID: 36275615), and a primary localization study confirms that HMGCR "is an integral membrane protein of the endoplasmic reticulum (ER)" (PMID: 38397481).

HMG-CoA is a bulky, charged coenzyme-A thioester; it does not freely cross the mitochondrial inner membrane, and no carrier has been described that exports mitochondrial matrix HMG-CoA to the cytosol/ER for mevalonate synthesis. Within the matrix, HMG-CoA is instead cleaved by HMG-CoA lyase to acetoacetate, committing it to ketone-body production. The compartment-confined nature of this flux is reinforced by the observation that renal HMGCS2-derived product acts locally and does not even contribute to the systemic circulating ketone pool (PMID: 41568909). This compartmentalization argument is decisive and answers the seed hypothesis's question about "context-dependent compartment coupling" in the negative: there is no described physiological route by which matrix HMG-CoA reaches the ER mevalonate machinery.



The two branches begin with the same metabolite in different compartments and never converge.

Finding 3 — Direct paralog comparison: mevalonate/isoprenoid BP terms are experimentally/orthology-supported on HMGCS1 but only IBA on HMGCS2

A side-by-side comparison of the rat paralogs' GO annotations is the clearest single piece of curation evidence and pinpoints the annotation-provenance problem:

Feature	HMGCS1 (rat, P17425)	HMGCS2 (rat, P22791)
Subcellular location	Cytoplasm / cytosol	Mitochondrion, mitochondrial matrix (GO:0005759, IDA:RGD)
Molecular function	GO:0004421 HMG-CoA synthase	GO:0004421 HMG-CoA synthase (IDA)
GO:0010142 (FPP biosynthesis, mevalonate)	Present, ISO:RGD (orthology, exp-backed)	Present, IBA:GO_Central only
GO:0019287 (isopentenyl-PP biosynthesis, mevalonate)	Present, ISO:RGD	Absent
GO:0033489 / 0033490 / 0036197 (cholesterol/sterol biosynthesis)	Present, ISO:RGD	Absent
Cellular response to cholesterol (GO:0071397)	Present, IEP:RGD	Absent
GO:0008299 (isoprenoid biosynthesis)	Consistent with pathway role	Not annotated
GO:0046951 (ketone body biosynthesis)	Absent	Present, IEP:RGD (core BP)

The pattern is unambiguous. On **HMGCS1**, the mevalonate/isoprenoid/sterol terms form a **coherent, mutually corroborating set** — orthology-supported ISO annotations for the biosynthetic steps plus a regulatory IEP term (cellular response to cholesterol) — exactly what one expects of a genuine pathway participant embedded in sterol-feedback regulation. On **HMGCS2**, only the single term GO:0010142 appears, and it appears with the **weakest evidence code in the set, IBA**, a phylogenetic propagation from the shared HMG-CoA synthase family node in GO_Central. HMGCS2 carries **none** of the accompanying sterol/isoprenoid biosynthesis terms and **none** of the cholesterol-response regulatory terms that would corroborate real

pathway membership. This asymmetry — a full, experimentally anchored program on HMGCS1 versus a lone, phylogenetically inferred term on HMGCS2 — is the diagnostic signature of paralog/ancestral carry-over rather than a genuine second function.

Mechanistic Model / Interpretation

The HMG-CoA synthase family has two vertebrate paralogs that arose by gene duplication and specialized to different subcellular compartments and metabolic fates:

1. **HMGCS1 (cytosolic)** condenses acetyl-CoA + acetoacetyl-CoA to HMG-CoA in the cytosol, where it feeds ER-membrane HMGCR and thence the mevalonate → farnesyl-PP → sterol/nonsterol isoprenoid pathway. Its regulation is sterol-responsive (SREBP-linked), consistent with a cholesterol/isoprenoid role.
2. **HMGCS2 (mitochondrial matrix)** performs the identical chemistry inside the mitochondrion, where the product HMG-CoA is committed to **ketogenesis** via HMG-CoA lyase. Its regulation is nutritional — fasting/PPAR α -driven and post-translationally tuned (e.g., SIRT5 desuccinylation at Lys367 shifts output toward acetoacetate; [PMID: 41580118](#)).

Because both enzymes catalyze the same reaction (GO:0004421) and produce the same metabolite, **automated phylogenetic annotation (GO_Central IBA) propagates the mevalonate-pathway BP term from the family's ancestral node onto both descendants**. This is a well-recognized source of paralog over-annotation. The molecular function is genuinely shared; the **downstream biological process is not**, because the physical fate of the shared metabolite is dictated by compartment and by the availability of the next enzyme in the pathway.

The seed hypothesis explicitly raises "possible context-dependent compartment coupling." The evidence does not support any such coupling under physiological conditions. Where HMGCS2 is reported to support mevalonate-type outputs — for example experimental complementation of HMGCS1-deficient systems, cytosolic-mistargeting constructs, or tumor overexpression — these represent **ectopic enzymatic capacity**: the enzyme *can* make HMG-CoA that, if placed in the correct compartment or expressed at supraphysiological levels, could seed mevalonate flux. That capacity is a property of the shared active site, not evidence of an endogenous HMGCS2 mevalonate function. GO biological-process annotations should reflect the *normal* physiological role, not the maximal in-vitro or overexpression potential.

The correct annotation profile for HMGCS2 is therefore:

- **MF:** GO:0004421 hydroxymethylglutaryl-CoA synthase activity — **retain** (IDA, shared, genuine).
 - **BP (core):** GO:0046951 ketone body biosynthetic process — **retain/emphasize**.
 - **CC:** GO:0005759 mitochondrial matrix (IDA:RGD) — **retain**.
 - **BP (mevalonate):** GO:0010142 — **remove or NOT-qualify** (IBA carry-over, compartment-incompatible).
 - **BP (isoprenoid):** GO:0008299 — **do not add**.
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Evidence Base

Citation	Evidence type	Supports/ Refutes/ Qualifies	Claim tested	Key finding	Context	Confid limitat
PMID: 41568909	Mutant phenotype / metabolomics	Refutes mevalonate; supports ketogenic	Is HMGCS2 the mitochondrial rate-limiting ketogenic enzyme?	"the rate- limiting enzyme for ketogenesis, mitochondrial HMGCS2, is induced in the proximal tubule ... during fasting"; product acts locally	Mouse/rat kidney proximal tubule	High fo ketogener
PMID: 25187565	Structural/ evolutionary (ortholog)	Competing (assigns mevalonate to paralog)	Which paralog performs the mevalonate committed step?	HMGS-1 is "the ortholog of human HMGCS1 enzyme, which mediates the first committed step of the mevalonate pathway"	<i>C. elegans</i> / human	High; c species explici
PMID: 36275615	Review	Qualifies (compartment barrier)	Where is the next mevalonate enzyme HMGCR?	HMGCR is "the rate limiting enzyme in the mevalonate pathway," "ER- localized"	Mammalian review	Medium (review mecha decisiv
PMID: 38397481	Localization (primary)	Qualifies/ Refutes (compartment barrier)	Is HMGCR accessible to matrix HMG- CoA?	HMGCR "is an integral membrane protein of the endoplasmic reticulum (ER)"	Human cells	High; establis physica separa

Citation	Evidence type	Supports/ Refutes/ Qualifies	Claim tested	Key finding	Context	Confid limitat
UniProt P22791 (RGD)	Localization / database	Refutes endogenous mevalonate role	HMGCS2 compartment & annotation basis	Mitochondrion/ matrix (GO:0005759 IDA); MF GO:0004421 IDA; GO:0010142 IBA only; GO:0008299 absent	Rat	High; c record
P17425 vs P22791 (paralog GO comparison)	Computational / database	Refutes (IBA carry-over pattern)	Evidence strength of GO:0010142 per paralog	HMGCS1 carries mevalonate/ sterol terms with ISO+IEP support; HMGCS2 carries only GO:0010142 via IBA and none of the corroborating terms	Rat	High; d proven contras
PMID: 40272888 / PMID: 39464122	Loss-of- function flux modeling	Supports ketogenic core role	HMGCS2 LOF phenotype	Disrupting HMGCS2, "the rate-limiting step of ketogenesis," impairs hepatic fat oxidation → MASLD-MASH phenotype; no sterol phenotype	Human + mouse liver	High

Citation	Evidence type	Supports/ Refutes/ Qualifies	Claim tested	Key finding	Context	Confid limitat
PMID: 41580118	Direct assay / PTM	Supports ketogenic core role	HMGCS2 product identity	HMGCS2 "the rate-limiting enzyme of ketogenesis"; SIRT5 desuccinylation (K367) shifts output to acetoacetate	Mouse kidney tubule	High; p is keton
PMID: 39808588	Mutant phenotype (knockdown)	Supports ketogenic core role	HMGCS2 endogenous output	Hepatic HMGCS2 knockdown reduces plasma ketones; "first reaction in ketogenesis"	Rat liver/ brain	High; k readou
PMID: 40288162	Expression / localization	Supports ketogenic core role	HMGCS2 tissue function	HMGCS2 "the rate-limiting enzyme of ketogenesis" highly expressed in chorionic plate; produces 3-HB	Human placenta	High fo ketoger

How the evidence coheres: Every retrievable primary study of HMGCS2 loss-of-function or expression reports a *ketogenic* readout (plasma ketones, hepatic fat oxidation, acetoacetate/ β -HB), never a sterol/isoprenoid deficit. The mevalonate role is consistently assigned to HMGCS1. The two mechanistic papers on HMGCR fix the downstream enzyme in the ER, closing the compartment loop. No primary study asserting endogenous HMGCS2-driven mevalonate/isoprenoid flux was retrievable.

GO Curation Implications (leads requiring curator verification)

GO term	Aspect	Current status on P22791	Recommended lead
GO:0004421 hydroxymethylglutaryl-CoA synthase activity	MF	IDA:RGD	Retain. Genuine, experimentally supported core molecular function shared with HMGCS1.
GO:0046951 ketone body biosynthetic process	BP	IEP:RGD	Retain / elevate as primary BP. Core physiological process.
GO:0010142 farnesyl diphosphate biosynthetic process, mevalonate pathway	BP	IBA:GO_Central	Remove or NOT-qualify. Compartment-incompatible paralog carry-over; belongs to HMGCS1.
GO:0008299 isoprenoid biosynthetic process	BP	Not annotated	Do not add. Over-broad and unsupported for the mitochondrial isoform.
GO:0005759 mitochondrial matrix	CC	IDA:RGD	Retain.

Curators should not "upgrade" the mevalonate/isoprenoid claim to an experimental or ISS annotation for HMGCS2. A NOT-qualifier on GO:0010142 (rather than silent deletion) is defensible given the recurrent nature of the IBA propagation, to prevent automated re-annotation. Note that "protein binding" is explicitly *not* recommended — a more informative and supported MF term (GO:0004421) exists.

Mechanistic Scope

The immediate molecular function under test is **HMG-CoA synthase activity** — condensation of acetyl-CoA + acetoacetyl-CoA to HMG-CoA + CoA. This chemistry is real and shared by both paralogs; it is not in dispute. The question is exclusively about the **biological process** to which HMGCS2's HMG-CoA is committed.

- **Direct gene-product activity (core):** production of HMG-CoA in the mitochondrial matrix, cleaved by HMG-CoA lyase to acetoacetate and thence to β -hydroxybutyrate — i.e., ketone-body synthesis.
- **Downstream / indirect effects (not core function):** systemic ketone signaling, hepatic fat-oxidation balance, MASLD-MASH protection, renal ischemia protection, tubule-macrophage inflammatory crosstalk, fetal ketone supply, and cognitive/mitochondrial effects. These are physiological consequences of ketogenesis, not evidence of mevalonate participation.
- **Ectopic capacity (not core function):** the ability of HMGCS2, when mistargeted or overexpressed, to generate HMG-CoA that could seed mevalonate flux. This is an intrinsic property of the shared active site, consistent with the seed's own "ectopic capacity only" framing.

The mevalonate/isoprenoid claim conflates a shared *molecular function* with a *compartment-specific biological process* — precisely the distinction the seed hypothesis was designed to probe.

Conflicts and Alternatives

- **Paralog confusion (primary alternative, favored):** GO:0010142 is propagated by IBA from the HMGCS ancestor shared with HMGCS1. The MF is common; the mevalonate BP is HMGCS1-specific. This is the most parsimonious explanation for the annotation.
- **UniProt free-text ambiguity:** The P22791 FUNCTION line notes that HMG-CoA can be "converted by HMG-CoA reductase (HMGCR) into mevalonate." Read literally this implies mevalonate participation, but it conflates the general chemistry of HMG-CoA with compartment-restricted physiology; in the mitochondrion, HMG-CoA is a ketogenic, not a mevalonate, intermediate. This text is a plausible source of the over-annotation and should not be used as primary support.

- **Ectopic capacity (compatible with the seed's own framing):** complementation or cancer/overexpression settings that place HMGCS2 activity where cytosolic HMG-CoA is usable demonstrate enzymatic *capacity*, not endogenous pathway participation.
 - **Organism/tissue differences:** ketogenic HMGCS2 expression is inducible and tissue-specific (liver, kidney, placenta); none of these contexts show a mevalonate output. No organism-specific exception was found.
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Limitations and Knowledge Gaps

1. **Annotation provenance verified from curated records, not re-run programmatically.** The IBA-vs-ISO evidence-code contrast was drawn from RGD/UniProt/GO_Central records. A curator should confirm the live evidence codes on P22791 and P17425 in QuickGO/RGD. This matters because the entire recommendation hinges on GO:0010142 being IBA-only on HMGCS2.
 2. **No direct isotopic flux tracing of mitochondrial HMG-CoA fate.** The compartment-barrier argument rests on the established ER localization of HMGCR and the absence of a matrix→cytosol HMG-CoA carrier, not on a dedicated tracing experiment. A positive tracer result is the only thing that could rescue the BP claim; its absence in the literature is consistent with refutation but is not a formal disproof.
 3. **Cancer-overexpression literature not exhaustively adjudicated.** Targeted PubMed searches for the seed's "complementation" and "cancer overexpression" scenarios returned no primary study asserting physiological mevalonate/isoprenoid flux by HMGCS2; where HMGCS2 appears in cancer, the retrievable primary literature ties it to ketogenesis/ketone-body signaling. The "ectopic capacity" scenario therefore remains a theoretical enzymatic-capacity argument and cannot be cited as physiological.
 4. **No local bioinformatics files were provided.** Analyses used public UniProt/RGD records and PubMed literature; no rat-specific structural or flux dataset was available in-run.
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Proposed Follow-up Experiments / Actions

1. **GO annotation provenance audit:** Programmatically pull QuickGO/RGD annotations for P22791 and P17425; confirm GO:0010142 is IBA-only on HMGCS2 and ISO/experimental on

HMGCS1, and that HMGCS2 lacks the sterol/isoprenoid corroborating terms. Apply a PAINT/GO_Central curator override restricting GO:0010142 to the HMGCS1 clade.

2. **Compartment-resolved isotope tracing:** In cells lacking HMGCS1 but expressing native (matrix-targeted) HMGCS2, trace ^{13}C -acetate/acetyl-CoA into cholesterol/FPP. Endogenous mevalonate participation predicts label incorporation; the ketogenic-only model predicts label appears instead in acetoacetate/ β -HB.
3. **Mistargeting rescue test:** Compare cytosol-retargeted HMGCS2 (mitochondrial-targeting-sequence deleted) versus native HMGCS2 for the ability to rescue mevalonate output in HMGCS1-null cells. Rescue only by the retargeted form confirms the function is compartment-gated capacity, not endogenous role.
4. **Loss-of-function readout specificity:** Re-examine existing HMGCS2-KO models for any sterol/isoprenoid deficit. Absence of such a phenotype (only ketogenic/fat-oxidation defects reported to date) further argues against a mevalonate role.
5. **Curation action:** Remove or NOT-qualify GO:0010142 on P22791; do not add GO:0008299; retain GO:0004421 (MF), GO:0046951 (core BP), and GO:0005759 (CC). Consider clarifying the UniProt FUNCTION free-text to avoid implying mitochondrial mevalonate synthesis.

Candidate references with exact snippets to verify: - [PMID: 41568909](#): "*the rate-limiting enzyme for ketogenesis, mitochondrial hydroxymethylglutaryl-CoA synthase 2 (HMGCS2), is induced in the proximal tubule of the kidney during fasting.*" - [PMID: 25187565](#): "*the ortholog of human HMGCS1 enzyme, which mediates the first committed step of the mevalonate pathway.*" - [PMID: 36275615](#): "*Endoplasmic reticulum (ER)-localized 3-hydroxy-3-methylglutaryl coenzyme A (HMG CoA) reductase (HMGCR), the rate limiting enzyme in the mevalonate pathway.*" - [PMID: 38397481](#): "*It is an integral membrane protein of the endoplasmic reticulum (ER).*"

Bottom Line

Rat HMGCS2 (P22791) is the mitochondrial-matrix ketogenic HMG-CoA synthase. Its endogenous biological process is ketone-body biosynthesis, not mevalonate/isoprenoid synthesis. The mevalonate function belongs to the cytosolic paralog HMGCS1, whose downstream partner HMGCR is ER-bound and physically inaccessible to matrix HMG-CoA. **GO:0010142 on HMGCS2 is an IBA paralog carry-over and should be removed or NOT-qualified; GO:0008299 should not be added. The shared molecular function (GO:0004421) and the ketogenesis BP / mitochondrial-matrix CC annotations should be retained.**

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